

AI-Powered Interview Training Platform

Problem Statement

Students and job seekers lack personalised, accessible tools to prepare for technical and behavioural interviews. Current platforms are static, recruiter-focused, and provide no real-time feedback or adaptive coaching.

- No real-time feedback on communication or confidence
- Mock interviews are generic, not role-specific
- No objective progress tracking over time
- Resume weaknesses unaddressed before applications
- Coding practice disconnected from interview simulation

Proposed Solution

An AI-driven self-paced training platform that simulates realistic interviews, analyses performance holistically, and generates personalised improvement plans based on the student's target role and readiness level.

- AI mock interviews adaptive to role & difficulty
- Real-time voice, video & confidence analysis
- Resume coaching with ATS gap detection
- Coding evaluator with AI-powered hints
- Interview Readiness Score & progress roadmap

Core Modules

Module	Description
Resume Coach	Parses resume, detects ATS gaps, suggests rewrites, and scores against a target job description.
Mock Interview AI	Generates role-specific questions (HR, Technical, Behavioural). Adapts difficulty based on answers.
Communication Analyser	Analyses speech pace, filler words, tone confidence, and facial expressions via audio/video AI.
Technical Evaluator	Domain coding challenges with AI code review, optimisation hints, and skill-gap heatmap.
Progress Dashboard	Interview Readiness Score (IRS/100), session history, improvement charts, and personalised roadmap.
Peer / AI Practice	Live peer mock-interview rooms and an AI interviewer that asks natural follow-up questions.

Technology Stack & Dataset Sources

Technology Stack	Dataset Sources
Frontend — React.js / Flutter	Resume — Kaggle Resume Dataset
Backend — FastAPI / Node.js	Interview — Interview QA Dataset
Database — MongoDB / Firebase	Emotion — FER2013 + RAVDESS
NLP — BERT / GPT fine-tune	Speech — Mozilla Common Voice
Speech AI — Whisper / Librosa	Coding — CodeSearchNet
Vision — OpenCV / CNN	NLP Train — SQuAD 2.0
Deployment — AWS / GCP	

Expected Outcomes

- Students arrive atrealinterviews with measurably higher confidence and preparation quality.
- Personalised AI feedback closes skill gaps faster than static prep resources.
- Resume quality improves before applications, increasing callback rates.
- Colleges can integrate the platform as a pre-placement training tool.
- Students in Tier-2/3 cities gain access to coaching quality previously available only via paid institutes.